

Pure Artery Malformation of Posterior Cerebral Artery with Dysplastic Internal Carotid Artery

Sir,

Pure arterial malformations (PAMs) are dilated, overlapping, and tortuous arteries with a coil-like appearance and/or mass of arterial loops without any associated venous component.^[1,2] The pathogenesis and etiology of PAMs are unclear, and they are thought to have a benign natural history and are usually managed conservatively.^[3] So far, only 39 cases have been reported in English language literature, out of which 7 were associated with aneurysm.^[4] The Brinjikji *et al.*^[3] have the largest series of 12 cases.

A 40-year-old female presented to the emergency ward with a history of sudden onset holocranial headache associated with episodes of vomiting for five days. There was no history of fever, seizure, or loss of consciousness. The patient's neurological evaluation was completely normal, and her vital parameters were within normal limits. A computed tomography scan (CT scan) of the brain was obtained, in which rounded hyper-densities were noted in bilateral parasellar regions [Figure 1a]. Upon further evaluation with digital subtraction angiography (DSA), dysplastic right cavernous and clinoidal segments of internal carotid artery (ICA) along with tortuous, redundant vascular compact loops of purely arterial mass (PAM) of left posterior cerebral artery (PCA) were evident [Figure 1b and c]. There was no evidence of nidus or early draining vein on DSA.

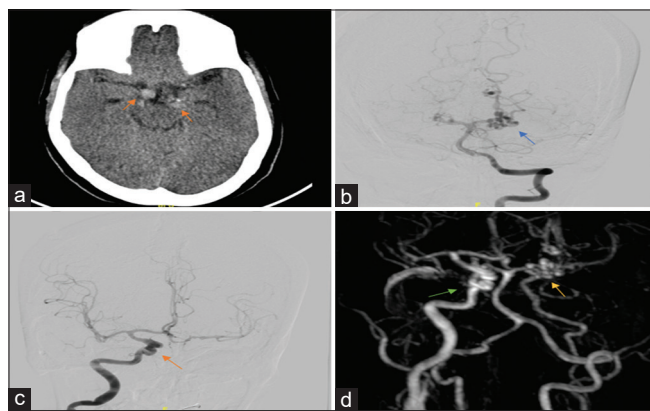


Figure 1: (a) Computed tomography of the brain axial section shows right parasellar and left crural cistern hyperdensities (orange arrows); (b) Digital subtraction angiography of left vertebral artery shows pure arterial malformation of the left posterior cerebral artery (blue arrow); (c) Digital subtraction angiography of the right internal carotid artery (ICA) shows dysplastic cavernous and clinoidal segment of right ICA (orange arrow); (d) Magnetic resonance angiography (MRA) shows dysplastic right ICA (green arrow) and pure arterial malformation of the left posterior cerebral artery (orange arrow)

Magnetic resonance angiography (MRA) also showed the dysplastic right ICA and PAM of left PCA [Figure 1d]. There was no evidence of parenchymal hemorrhage, infarction, or subarachnoid hemorrhage (SAH) on CT or magnetic resonance imaging (MRI) of the brain.

Investigations for vasculopathy and infective pathology were also negative. The patient was managed conservatively and advised for regular radiological follow-up.

The etiology, pathogenesis, and natural history of PAM remain unclear. Due to the rarity of this entity, there is no consensus on optimal treatment. However, Brinjikji *et al.*^[3] suggested that patients with PAM should be treated conservatively. Liu *et al.*^[4] further added that conservative management is useful for PAM without local aneurysm, while a patient with PAM with a local aneurysm can achieve a better prognosis when treated via intervention or endovascular treatment. They may be mistaken as aneurysm or arteriovenous malformations on non-invasive imaging.^[3] The absence of nidus and venous components makes the PAM distinct from arteriovenous malformations or fistulas. The diagnosis of this entity requires DSA and 3D imaging examinations.^[4]

We, with this, present one of the handfuls of pure arterial malformation cases to add to the limited database for enhancing future study or treatment guidelines.

Declaration of patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form the patient(s) has/have given his/her/their consent for his/her/their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.

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Conflicts of Interest

There are no conflicts of interest.

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